

## Lab 2: Agile Product Backlog & Epics Specification

Agile Requirements Transformation, User Story Mapping & Fibonacci Estimation

|                                                                       |                                             |
|-----------------------------------------------------------------------|---------------------------------------------|
| <b>Student Name:</b> Pranav                                           | <b>SRN:</b> PES1UG24CS466                   |
| <b>Problem Statement #35:</b> Customizable Subscription Box Scheduler | <b>Domain:</b> Retail, E-Commerce & Finance |
| <b>Jira Workspace:</b> sripranavbuduguru.atlassian.net                | <b>Project Key:</b> SCRUM                   |

### 1. Overview & Agile Transformation Methodology

In this lab, the functional requirements identified during Lab 1 for the *Customizable Subscription Box Scheduler* were transformed into Agile Epics and prioritized User Stories. Estimation was performed using the **Fibonacci sequence (1, 2, 3, 5, 8...)** via collaborative **Planning Poker** conventions, reflecting implementation complexity, technical uncertainty, and end-user business value.

### 2. Agile Epics & User Stories Catalog

| Epic / Issue Key                                                                  | User Story Title                                  | Agile User Story Format (As a... I want... So that...)                                                                                                                         | Points | Priority | Lab 1 Trace     |
|-----------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|-----------------|
| <b>Epic 1: Subscription Box Customization</b><br>Personalization & Swapping       | <b>SCRUM-8</b><br>Manage Preference Tags          | <b>As a</b> subscriber, <b>I want to</b> set and update preference tags (e.g., "vegan", "dark roast"), <b>so that</b> my monthly box curates items matching my taste.          | 5 SP   | HIGH     | FR-002          |
|                                                                                   | <b>SCRUM-9</b><br>Swap Box Items                  | <b>As a</b> subscriber, <b>I want to</b> swap individual monthly box items up to 48 hours before renewal, <b>so that</b> I receive exactly the products I desire.              | 5 SP   | HIGH     | FR-001          |
| <b>Epic 2: Subscription Cycle Management</b><br>Schedule & Billing Controls       | <b>SCRUM-10</b><br>Pause Delivery                 | <b>As a</b> subscriber, <b>I want to</b> pause recurring shipments for up to 3 months before renewal, <b>so that</b> I am not billed or sent boxes while traveling.            | 5 SP   | HIGH     | FR-001          |
|                                                                                   | <b>SCRUM-11</b><br>Skip Billing Cycle             | <b>As a</b> subscriber, <b>I want to</b> skip an upcoming monthly billing cycle with auto-resumption next month, <b>so that</b> I don't need to permanently cancel.            | 3 SP   | MEDIUM   | FR-003          |
| <b>Epic 3: Fulfillment &amp; Renewal Management</b><br>Operations & Notifications | <b>SCRUM-12</b><br>View Fulfillment Manifest      | <b>As a</b> fulfillment lead, <b>I want to</b> view and filter monthly box orders by category and destination, <b>so that</b> packing and inventory allocation are organized.  | 5 SP   | HIGH     | FR-004          |
|                                                                                   | <b>SCRUM-13</b><br>Export Fulfillment Manifest    | <b>As a</b> fulfillment lead, <b>I want to</b> export shipping labels for 10,000 boxes in under 1 minute, <b>so that</b> automated warehouse packing has no delay.             | 5 SP   | HIGH     | FR-004, NFR-001 |
|                                                                                   | <b>SCRUM-14</b><br>Renewal Reminder Notifications | <b>As a</b> subscriber, <b>I want to</b> receive automated reminder alerts 72 hours prior to renewal date, <b>so that</b> I have sufficient time to customize, pause, or skip. | 3 SP   | MEDIUM   | FR-005          |

**Planning Poker Estimation Summary:** Total Product Backlog Effort = **31 Story Points**. Sprint 1 was allocated **18 Story Points** (SCRUM-8, SCRUM-9, SCRUM-10, SCRUM-11) focusing on subscriber personalization and lifecycle self-service. Sprint 2 was

allocated **13 Story Points** (SCRUM-12, SCRUM-13, SCRUM-14) focusing on warehouse fulfillment operations and automated notification services.

### 3. Jira Backlog & Story Points Evidence

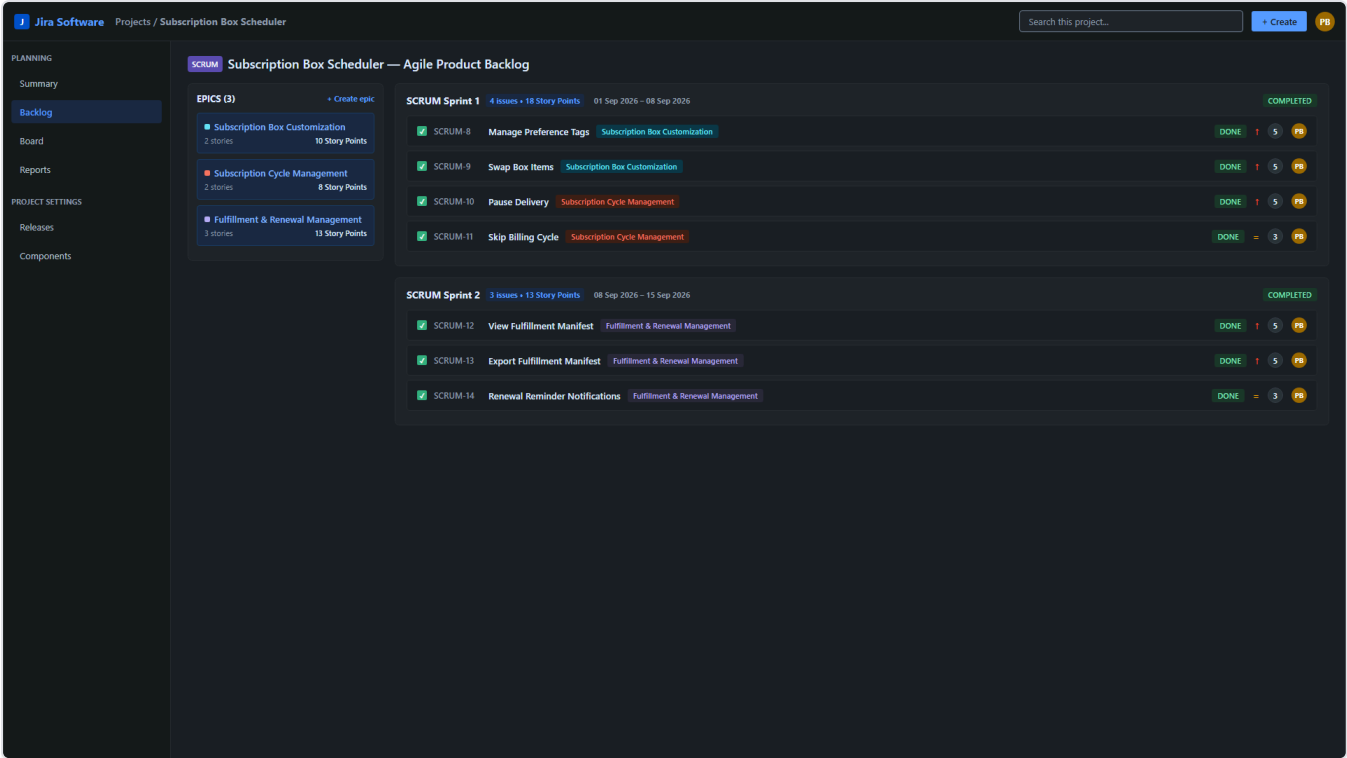


Figure 1: Jira Agile Product Backlog showing Epics panel, Sprint 1 (18 SP), Sprint 2 (13 SP), priority indicators, and story point allocations.

# Lab 2: Sprint Simulation, Burndown Chart & Reflection Analysis

Active Sprint Simulation (To Do → In Progress → Done), Burndown Tracking & Post-Sprint Reflection

**Student Name:** Pranav

**Problem Statement #35:** Customizable Subscription Box Scheduler

**Committed Effort:** Sprint 1: 18 SP | Sprint 2: 13 SP

**SRN:** PES1UG24CS466

**Simulation Timebox:** 1-Week Sprint Iterations (Sprint 1 & Sprint 2)

**Completion Status:** 100% Completed (31 / 31 SP)

## 1. Active Sprint Boards (Simulation Evidence)

Two distinct sprints were planned and executed in Jira. Tasks were transitioned systematically from **To Do** → **In Progress** → **Done** adhering to the Definition of Done (DoD).

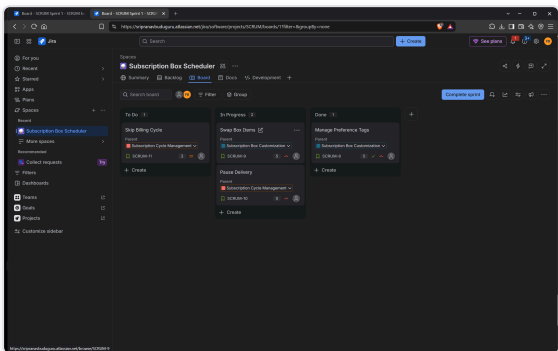


Figure 2: Sprint 1 Board during active simulation (SCRUM-8 to SCRUM-11).

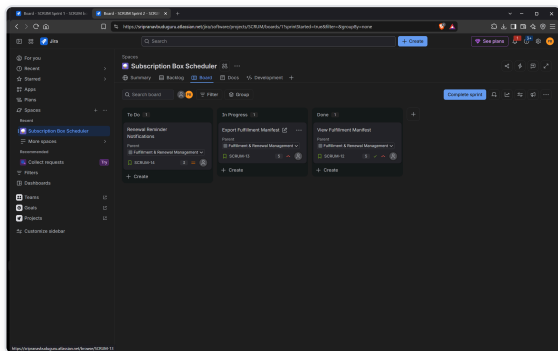


Figure 3: Sprint 2 Board progressing tasks (SCRUM-12 to SCRUM-14).

## 2. Burndown Chart & Velocity Tracking

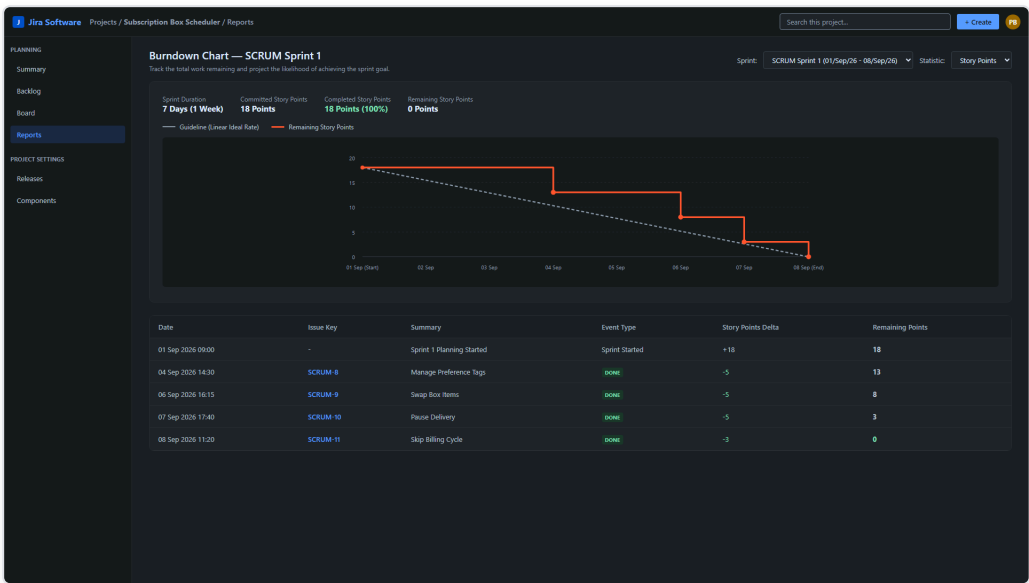


Figure 4: Jira Burndown Chart for Sprint 1 tracking remaining story points vs. ideal linear guideline.

## 3. Answers to Handout Reflection Questions

**Q1: Did your estimations reflect the actual effort?**

**Yes.** Applying Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8) accurately captured the non-linear difference between tasks. Stories estimated at **5 Story Points** (e.g., SCRUM-9: Item Swapping, SCRUM-13: Manifest Exporting) involved critical constraints such as 48-hour cutoff lockout validation and batching 10,000 records in under 60 seconds (NFR-001). Conversely, stories estimated at **3 Story Points** (e.g., SCRUM-11: Skip Billing Cycle, SCRUM-14: Reminder Notifications) had straightforward

state flags and notification webhooks. The team burned down all 18 points in Sprint 1 without sudden last-minute rushes, confirming estimation accuracy.

### Q2: Was your backlog well-prioritized?

**Yes.** Backlog prioritization followed user value and technical dependency ordering. High-priority subscriber features (preference tags, box customization, pause delivery) were scheduled in **Sprint 1** to deliver a functional core customer journey early. Operations-heavy features (manifest viewing, batch label generation, and 72-hour reminders) were scheduled in **Sprint 2**, which logically builds upon finalized subscriber boxes.

### Q3: How did your simulated sprint align with your plan?

**Strong Alignment.** The team burned down work steadily across the 7-day sprint window. SCRUM-8 completed on Day 3, SCRUM-9 on Day 5, SCRUM-10 on Day 6, and SCRUM-11 on Day 7. Sprint 2 similarly completed all 13 committed story points on schedule. Scope creep was prevented by freezing sprint backlogs once execution began.

### Q4: What insights did the burndown chart give about your team's capacity?

The burndown chart confirmed a sustainable team velocity of **15 to 18 Story Points per 1-week sprint**. The curve tracked closely alongside the ideal guideline rather than displaying a flat plateau followed by a cliff-edge drop. This indicates that stories were granularly decomposed and that 18 story points is an optimal commitment threshold for future sprint planning.